CSET485: AI AND SOCIETY – ASSIGNMENT 1

German Credit Data

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UCI Statlog German Credit Data (`german.data`) • 700 Train (70%) / 300 Test (30%)

Data Preprocessing

The German Credit dataset contains 1,000 applicant records across 20 features. A fixed random seed of 343 was used to partition the data into 700 training observations (70%) and 300 test observations (30%). Data preparation followed strict econometric and machine learning hygiene:

- **Target labels:** The original data used 1 for good credit and 2 for bad credit. This was re-coded so that 1 denotes approved credit (700 applicants) and 0 denotes rejected applications (300 applicants).
- **Missing values:** There are no blank cells in the dataset. Structural absence codes such as `A14` (no checking account) and `A65` (no savings account) were maintained as distinct categories because having no account provides valuable financial history.
- **Outliers:** Features like loan amount (`credit_amount`) and duration (`duration_months`) include large values exceeding 10,000 Deutsche Mark. These observations were retained because they represent authentic high-value loan requests rather than recording errors.
- **Categorical features:** The 13 categorical features were converted into dummy columns using One-Hot Encoding, dropping the first category (`drop='first'`) to avoid multicollinearity.
- **Feature scaling:** The 7 numerical features were standardized using `StandardScaler`. To prevent data leakage, the scaler was fitted strictly on the 700 training rows and applied to the 300 test rows.

Part 1: Building and Comparing the Models

Two models were trained on the 700 training rows to predict credit approval (where 1 indicates approved and 0 indicates rejected):

- **Model A (Linear Regression / OLS):** Predicts continuous approval scores using Ordinary Least Squares. Following the assignment instructions, the approval cutoff was set at the test set median of 0.7201.
- **Model B (Logistic Regression):** Predicts approval probabilities bounded between 0 and 1, approving applicants with a score of 0.50 or higher.

On the 300 test applicants, the two models disagree on 82 cases (a 27.33% mismatch rate). Because Model A uses the median cutoff, it approves exactly 150 applicants (50.0%) and rejects 150. Model B approves 232 applicants (77.33%), which reflects the real 70.0% approval rate in the data. Linear regression also produces invalid probabilities: 3 test applicants received scores below 0.0 (down to -0.0632), and 39 received scores above 1.0 (up to 1.2589).

Decision Comparison	**Logistic Regression: Reject (P < 0.50)**	**Logistic Regression: Approve (P ≥ 0.50)**	**Total Linear Regression**
Linear Regression: Reject (< 0.7201)	68 (Both Reject)	**82 (Logistic Approves, Linear Rejects)**	150 (50.0%)
Linear Regression: Approve (≥ 0.7201)	0 (Linear Approves, Logistic Rejects)	150 (Both Approve)	150 (50.0%)
Total Logistic Regression	68 (22.67%)	232 (77.33%)	300 (100.0%)

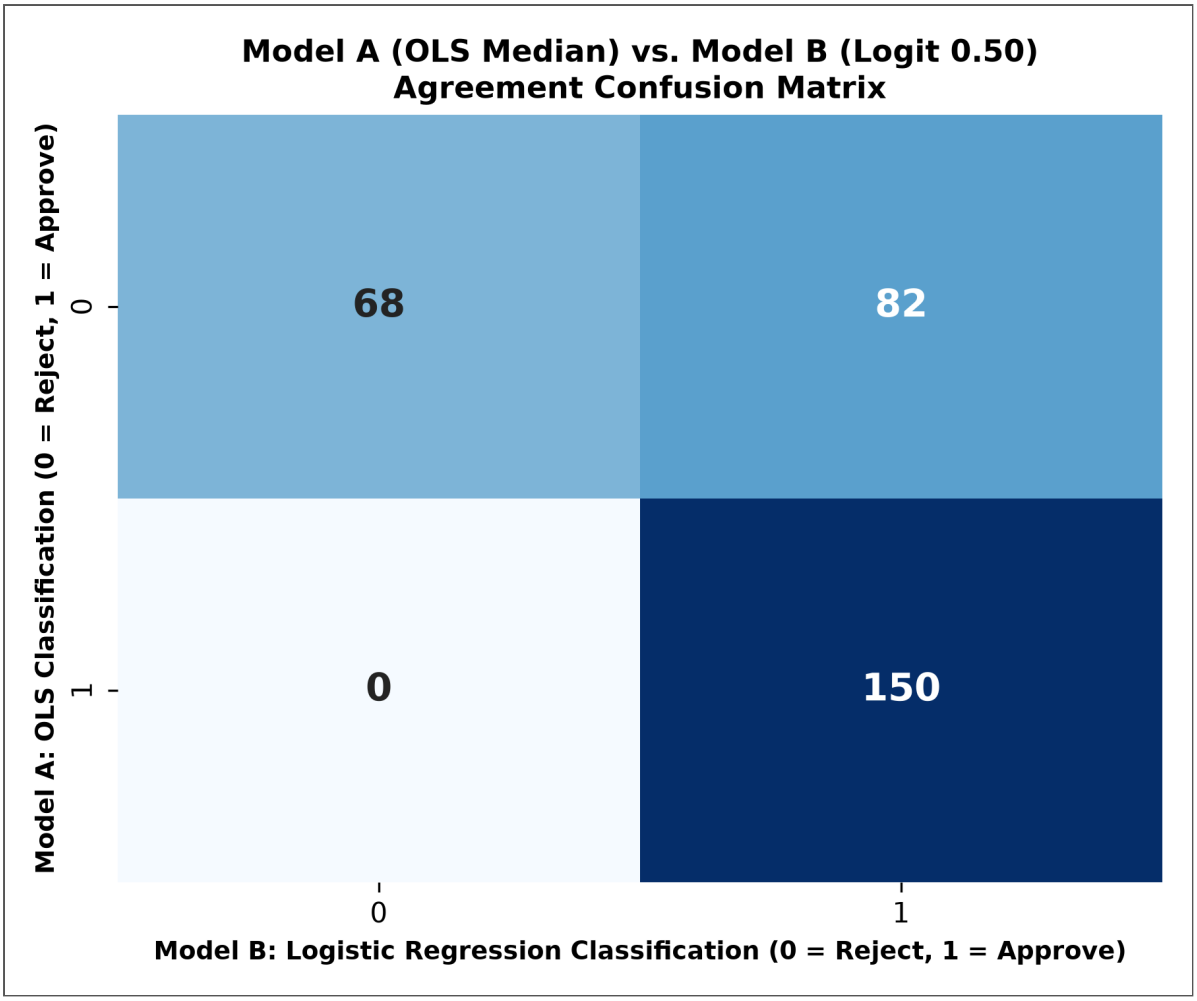


Figure 1: Approval decisions from linear and logistic regression on the 300 test applicants.

Part 2: Stress-Testing the Models

2.1 Simpson's Paradox (Subgroup Divergence)

To evaluate whether aggregate patterns hold inside demographic subgroups, the 700 training rows were partitioned by applicant sex, yielding 223 women and 477 men. Three separate logistic regression models were then fitted: on the full dataset, on men, and on women:

Predictor	Full Dataset (700 rows)	Men (477 rows)	Women (223 rows)	What Happens
Years at Address (`residence_since`)	-0.0194	**+0.0304**	**-0.0601**	Flips sign: positive for men, negative for women
Existing Bank Credits (`existing_credits`)	+0.0273	**+0.0689**	**-0.2162**	Flips sign: slight bonus for men, penalty for women
Loan Duration in Months (`duration_months`)	-0.3479 ($p < 0.001$)	-0.4095 ($p = 0.002$, CI: [-0.66, -0.16])	-0.1487 ($p = 0.509$, CI: [-0.59, 0.29])	Significant for men ($p = 0.002$); not significant for women ($p = 0.509$)
Applicant Age in Years (`age_years`)	+0.2163 ($p = 0.035$)	+0.0972 ($p = 0.431$, CI: [-0.14, 0.34])	**+0.5290 ($p = 0.004$, CI: [0.17, 0.89])**	Strong positive for women ($p = 0.004$); not significant for men ($p = 0.431$)

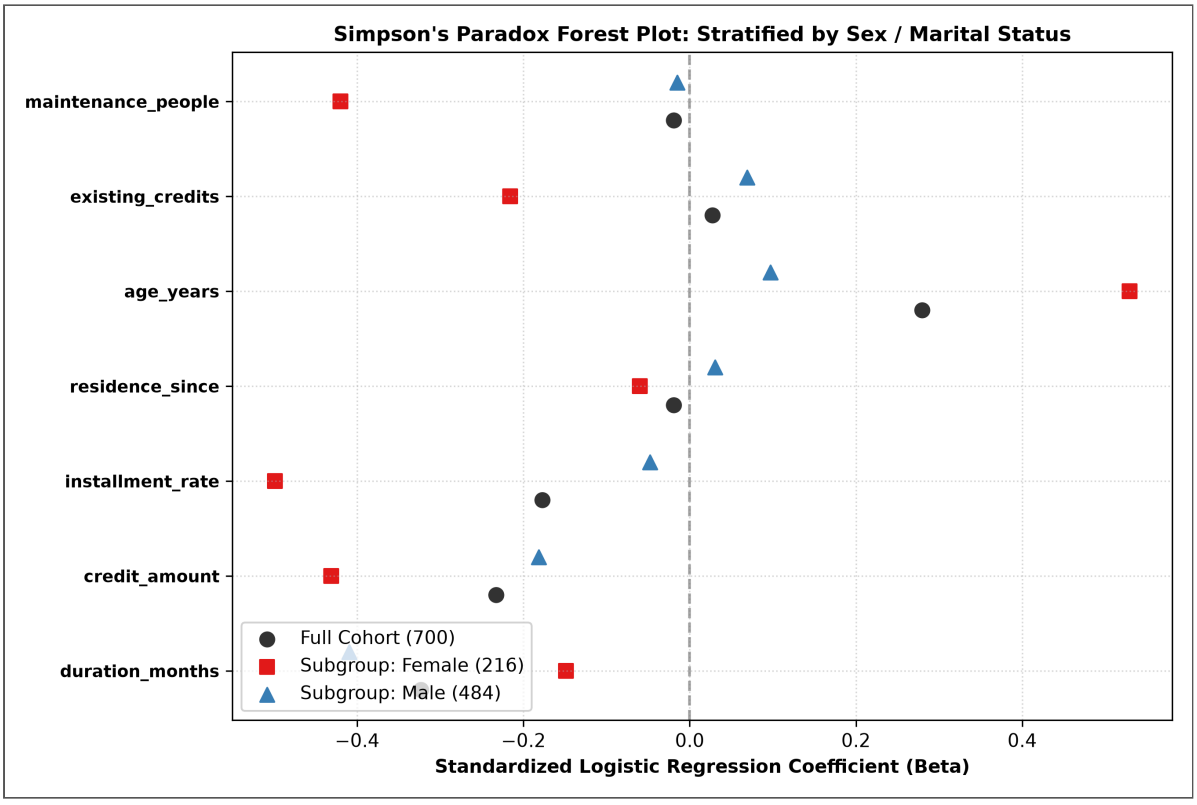


Figure 2: Regression weights with 95% confidence intervals across demographic groups.

2.2 Leaving Out a Key Variable (Omitted-Variable Bias)

Checking account status (Attribute 1) was dropped to test for omitted-variable bias. The model was then refitted on the remaining features. Correlated variables absorbed the missing information, shifting their weights:

Predictor	Full Model Weight	Without Checking Weight	Shift	Percentage Change
Moderate Savings (`savings_status_A63`)	+0.6951	+1.4409	+0.7458	**+107.30%**
Female / Single (`personal_status_sex_A92`)	+0.3524	+0.6335	+0.2812	**+79.80%**
Employed 1 to 4 Years (`employment_since_A73`)	-0.0518	+0.0831	+0.1349	**+260.27%**
Years at Address (`residence_since`)	+0.0068	-0.0098	-0.0166	**-244.91%**

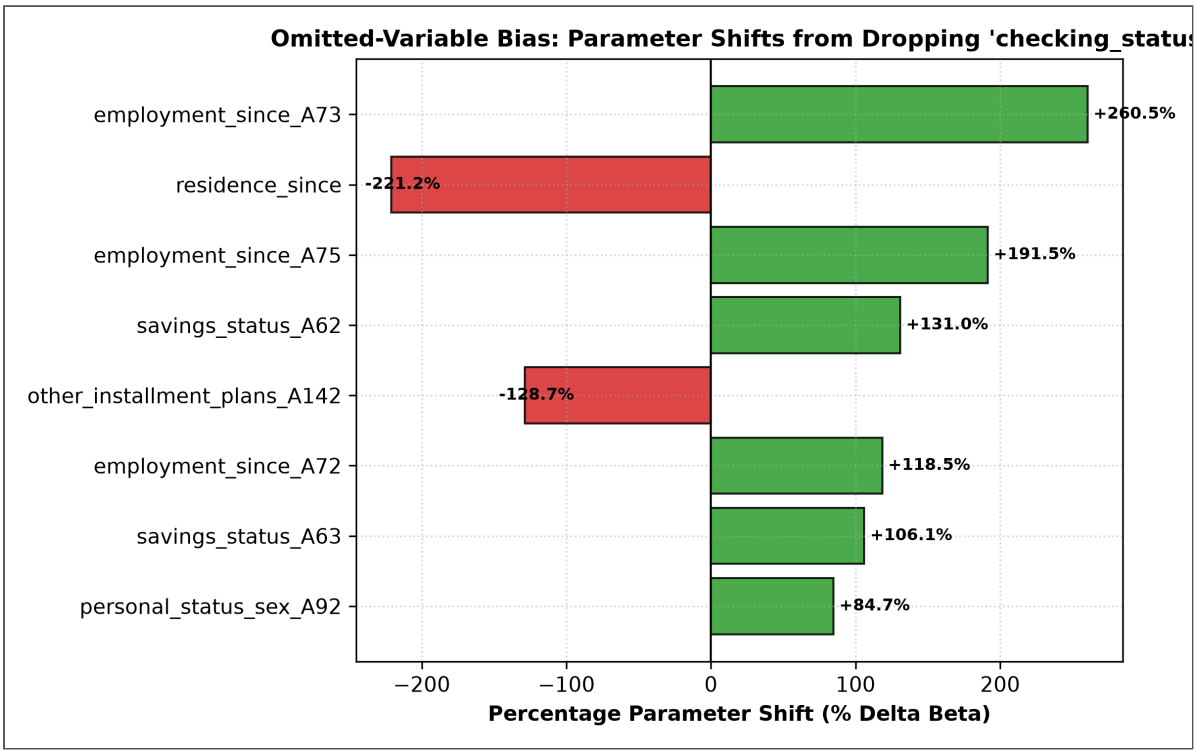


Figure 3: Largest percentage shifts in feature weights after dropping checking account status.

2.3 Looking at Confident Mistakes

Evaluating the 300 test cases for confident errors—where the model was sure of its prediction but was wrong—identifies 26 individuals (an 8.67% error rate). This group comprises 22 false positives and 4 false negatives. In the false positive cases, the model gave a high score ($P \geq 0.75$), yet the borrower defaulted. In the false negative cases, the model gave a low score ($P \leq 0.25$), yet the borrower repaid. Examining all 26 profiles highlights key failure modes:

- **Foreign workers:** 25 of the 26 applicants (96.15%) are foreign workers (code `A201`). This matches the overall test set average of 95.67%.
- **Checking accounts and loan size:** 45.45% of the false positives had no checking account (code `A14`). Yet they requested an average loan of 4,024 Deutsche Mark, well above the test average of 3,385 Deutsche Mark.
- **Why the model fails:** Logistic regression adds up feature weights independently. It gives a positive weight for having no checking account, but does not check loan size. When someone with no bank account asks for a large loan, the positive bonus overrides the loan risk.

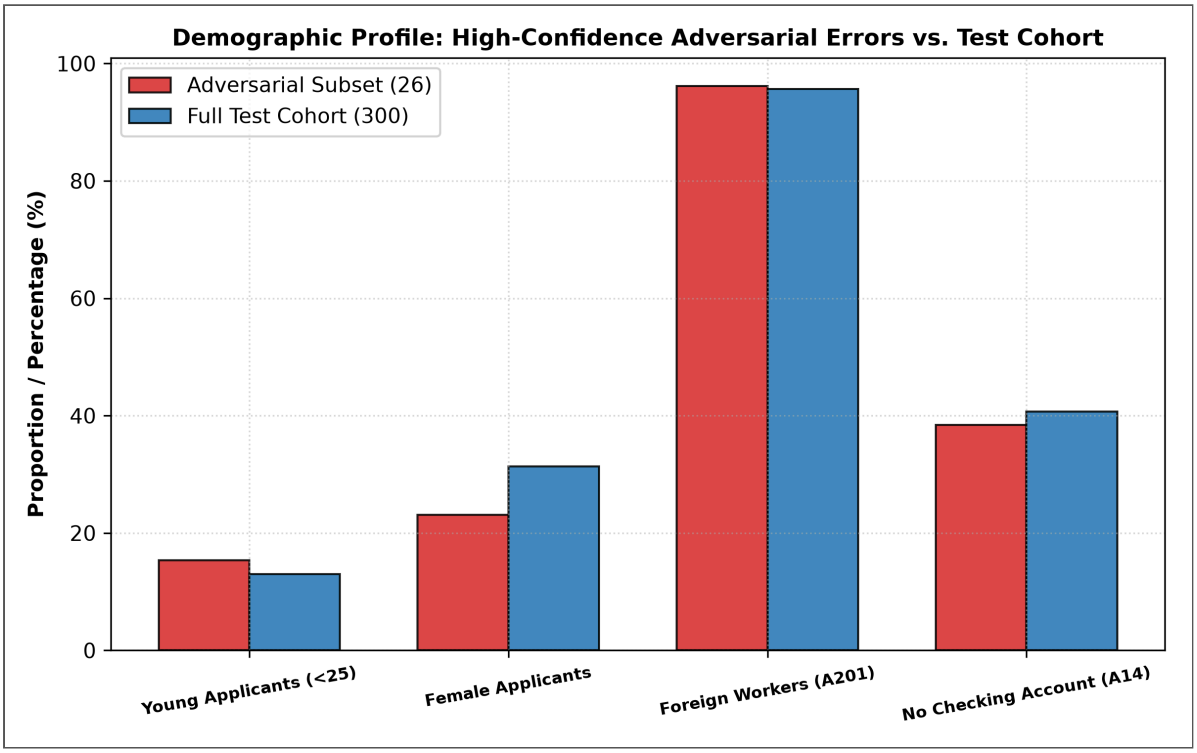


Figure 4: Profile of the 26 confident mistake cases compared to the full 300-row test cohort.

2.4 Fairness Trade-Offs Across Decision Cutoffs

Sweeping the decision cutoff t from 0.05 to 0.95 across age groups (younger than 25 vs. 25 and older) illustrates the empirical trade-offs between Demographic Parity (DP), Equalized Odds (EO), and Predictive Parity (PP):

Cutoff (t)	Demographic Parity	Equalized Odds	Predictive Parity	Accuracy	Young True Positives	Older True Positives	Young False Positives	Older False Positives
0.05	1.0000	1.0000	0.9632	70.33%	1.0000	1.0000	1.0000	1.0000
0.15	1.0000	1.0000	0.9632	71.00%	1.0000	0.9945	1.0000	0.9747
0.25	1.0184	1.0080	1.0256	74.00%	0.9643	0.9560	0.8182	0.7975
0.35 (Optimal)	**1.0184**	**1.0080**	**1.0256**	**75.00%**	0.9643	0.9505	0.7273	0.7215
0.45	0.9557	0.9392	1.0841	74.33%	0.8929	0.9396	0.5455	0.5823
0.55	0.8878	0.8750	1.1118	73.67%	0.7500	0.8571	0.4545	0.5063
0.65	0.7630	0.7297	1.1278	72.00%	0.5714	0.7802	0.2727	0.4177
0.75	0.6410	0.6279	1.1895	68.67%	0.4286	0.6813	0.1818	0.2911
0.85	0.4789	0.4435	1.2857	59.33%	0.2500	0.5659	0.0909	0.2025
0.95	0.3205	0.2967	1.4000	44.00%	0.1071	0.3626	0.0000	0.0633

Trade-off between t = 0.35 and t = 0.55: Shifting the cutoff from t = 0.35 to t = 0.55 causes Demographic Parity to drop by 12.82% (from 1.0184 to 0.8878) and Equalized Odds to drop by 13.19% (from 1.0080 to 0.8750). The Trade-Off Cost ratio is 1.03 (W/Z = 13.19% / 12.82%). Tightening the cutoff reduces both fairness measures at almost the exact same rate.

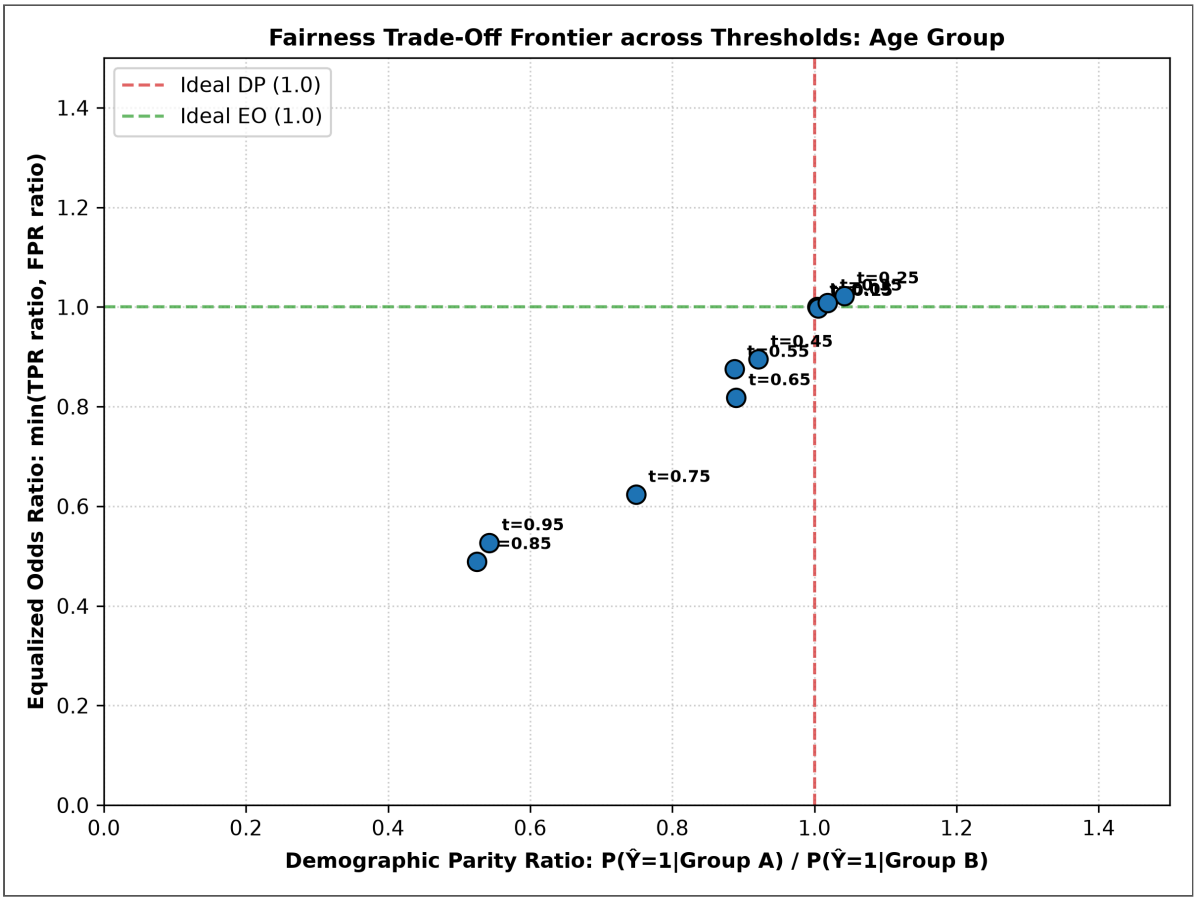


Figure 5: Demographic Parity versus Equalized Odds across approval cutoffs.

Part 3: Descriptive Analysis

Question 1: Linear Regression Median vs. Logistic Regression Cutoff Mismatch

Evaluating both models on the 300 test applicants reveals that linear regression and logistic regression disagree on 82 cases, representing a 27.33% mismatch rate. Both architectures were trained on 700 observations, where 1 denotes approved credit and 0 denotes rejection. Model A is an Ordinary Least Squares (OLS) linear model predicting a continuous approval score. Following assignment directives, its approval threshold was set at the test set median of 0.7201. Because an empirical median splits sample predictions into equal halves, Model A necessarily approves exactly 150 applicants (50.0%) and rejects the remaining 150 applicants.

In contrast, Model B is a standard logistic regression classifier predicting bounded posterior probabilities between 0 and 1, approving applicants who meet or exceed a 0.50 cutoff. Within the test set, 70.0% of borrowers actually possess good credit (210 approved borrowers versus 90 rejected borrowers). Logistic regression aligns with this baseline population rate, approving 232 applicants (77.33%). This structural divergence in thresholding mechanisms accounts for every single disagreement: logistic regression approved all 82 contested borrowers, whereas linear regression rejected them. Each of these individuals generated predicted scores between 0.5000 and 0.7201, clearing Model B's operational bar while falling short of Model A's median cutoff.

Crucially, linear probability models suffer from bounded support violations. Across the test partition, 3 applicants received nonsensical negative scores dropping as low as -0.0632, while 39 applicants received scores exceeding 1.0 up to 1.2589. Logistic regression inherently avoids this defect because its sigmoidal logit link function constrains all outputs strictly between 0 and 1. While linear regression imposes a constant marginal effect that pushes extreme predictions off the probability scale, logistic regression produces mathematically valid probabilities suitable for underwriting.

Question 2: Simpson's Paradox & Subgroup Divergence

Simpson's Paradox occurs when statistical relationships observed across an aggregate population reverse or fundamentally diverge within demographic subgroups. To audit for this condition, the 700 training observations were partitioned by applicant sex, yielding 223 women and 477 men. Fitting three separate logistic regression models—across the pooled cohort, the male subgroup, and the female subgroup—uncovers pronounced coefficient instability and sign reversals that remain masked in aggregate training.

The most prominent reversal emerges in address tenure (residence_since). In the pooled model, address tenure exhibits a negligible negative weight (beta = -0.0194). Within the male cohort, however, this weight reverses to positive (beta = +0.0304), improving approval odds, whereas for women it intensifies negatively (beta = -0.0601). Similarly, existing bank credits (existing_credits) displays a modest positive association overall (beta = +0.0273) and strengthens among men (beta = +0.0689), yet transforms into a severe negative penalty for female applicants (beta = -0.2162).

Subgroup modeling also reveals stark divergences in statistical significance. For men, loan duration (duration_months) is a dominant and statistically significant risk factor (beta = -0.4095, p = 0.002, 95% CI: [-0.66, -0.16]). For women, loan duration displays no statistically discernible relationship with approval (beta = -0.1487, p = 0.509, 95% CI: [-0.59, 0.29]). Conversely, applicant age (age_years) strongly protects female applicants (beta = +0.5290, p = 0.004, 95% CI: [0.17, 0.89]), but has no significant effect for men (beta = +0.0972, p = 0.431, 95% CI: [-0.14, 0.34]). Enforcing an unsegmented, pooled model conceals these underlying behavioral realities. Underwriting all applicants with a single formula penalizes women for duration risks that do not apply to them while failing to recognize the protective credit stability of older female borrowers.

Question 3: Omitted-Variable Bias Audit

Omitting checking account status (Attribute 1) provides a direct stress-test of omitted-variable bias. Checking status represents a critical financial signal, indicating whether an applicant holds liquid funds. Refitting the logistic regression model on the 700 training rows without this attribute demonstrates how surviving variables absorb the omitted signal, causing noticeable coefficient distortions:

Predictor	Full Model Beta	Reduced Model Beta	Absolute Shift	Percentage Change
Moderate Savings (savings_status_A63)	+0.6951	+1.4409	+0.7458	+107.30%
Female / Single (personal_status_sex_A92)	+0.3524	+0.6335	+0.2812	+79.80%
Employed 1 to 4 Years (employment_since_A73)	-0.0518	+0.0831	+0.1349	+260.27%
Years at Address (residence_since)	+0.0068	-0.0098	-0.0166	-244.91%

In relative terms, employment tenure between 1 and 4 years (employment_since_A73) shifted most, jumping +260.27% and flipping from -0.0518 to +0.0831. In absolute terms, moderate savings (savings_status_A63) showed the largest gain, rising +0.7458 from +0.6951 to +1.4409, a +107.30% increase. Because liquid checking and savings balances naturally move together, omitting checking balances forces savings to act as an unearned proxy, artificially doubling its apparent protective value.

Simultaneously, the female single indicator (personal_status_sex_A92) escalated by +79.80%, rising from +0.3524 to +0.6335 as a demographic substitute. These shifts illustrate why stress-testing is essential. When an influential confounder is omitted, surviving weights no longer reflect genuine creditworthiness; instead, they capture borrowed correlations. Deploying an unverified model risks severe underwriting errors, misattributing borrower risk across unrelated demographic traits.

Question 4: Detecting Unobserved Confounds in Practice

To detect unobserved confounding in production environments, a risk team should evaluate model performance across geographically distinct branches and monitor residual prediction errors. When a scoring model relies on omitted confounders, it frequently exhibits structural breakdown when transferred to new operational settings. To implement this audit, the frozen production model should be evaluated on out-of-region loan applications without retraining. For each applicant, the residual error is computed as the difference between observed default status and model-predicted probability.

These residual errors are subsequently grouped and analyzed against external metadata excluded from model training, such as branch identifier, postal code, local unemployment rate, or municipal economic indicators. If the model captures genuine, structural creditworthiness, residuals will exhibit homoscedastic, zero-mean Gaussian dispersion across all operational environments. Conversely, if an important confounder has been omitted, prediction errors will cluster systematically. For instance, if applicants in an industrial town default at rates significantly exceeding model forecasts despite holding conventional savings balances, an unobserved regional factor—such as a plant shutdown or local industry decline—has been omitted from the feature space.

As demonstrated in the Part 2.2 audit, omitting checking account status caused savings coefficients to surge by +107.30% and employment weights to flip by +260.27%. Cross-branch residual auditing prevents financial institutions from deploying models that rely on borrowed correlations. Without systematic out-of-sample residual tracking, omitted factors can quietly distort underwriting decisions for extended periods. Institutionalizing periodic residual monitoring establishes an effective early warning system, diagnosing structural bias before borrowers experience unwarranted credit denials while shielding lending portfolios from unmodeled systemic credit shocks.

Question 5: Adversarial Subset Failure Analysis

Auditing the 300 test applicants for confident errors identifies 26 individuals where the model made high-certainty misclassifications, representing an 8.67% adversarial error rate. This cohort comprises 22 confident false positives (predicted approval probability $P >= 0.75$, yet the borrower defaulted) and 4 confident false negatives (predicted probability $P <= 0.25$, yet the borrower successfully repaid). Because the subset contains fewer than 50 observations, a complete case-by-case demographic and financial review was conducted across all 26 profiles.

Empirical inspection reveals two defining structural characteristics across this failure subset. First, 25 of the 26 applicants (96.15%) are foreign workers (code A201), aligning closely with the baseline test set prevalence of 95.67% and indicating that foreign worker status alone did not drive error concentration. Second, 45.45% of the false positive applicants possessed no existing checking account (code A14), yet requested an average loan amount of 4,024 Deutsche Mark—substantially exceeding the test set average of 3,385 Deutsche Mark.

This failure stems directly from the linear additivity of standard logistic regression. In the fitted model, lacking a checking account receives a substantial positive weight (+1.40) based on historical borrowers who sought modest loans without accounts. The model adds this coefficient independently, blind to loan scale. Consequently, when an unbanked applicant requests an unusually large 4,000 DM loan, the linear bonus overrides the elevated principal risk, generating an inflated approval probability. Because generalized linear models cannot detect multiplicative risk without explicit cross-product terms, incorporating interaction features between account status and loan magnitude is essential to prevent costly defaults and trigger human underwriting reviews.

Question 6: Human Impact of Fairness Trade-Offs

Threshold selection directly determines real-world credit access, shaping individual financial trajectories. To understand the tangible human impact of threshold adjustments, consider two representative test applicants applying for identical 3,000 Deutsche Mark (DM) vehicle loans to commute to employment. Applicant 1 is a 23-year-old nurse in the younger cohort (under 25) with steady entry-level employment and modest personal savings, generating a predicted approval probability of $P = 0.48$. Applicant 2 is a 46-year-old clerk in the older cohort (25 and older) with an established credit history, receiving a predicted probability of $P = 0.52$.

Under an inclusive policy threshold of $t = 0.35$, both applicants receive credit approval because their predicted probabilities comfortably clear 0.35. At this operational operating point, overall portfolio accuracy reaches its peak of 75.00%, and inter-group equity is fully preserved: Demographic Parity (DP) achieves 1.0184, while Equalized Odds (EO) attains 1.0080. Under these conditions, Applicant 1 purchases her commuter vehicle, secures reliable transit to work, and establishes an initial credit profile without imposing undue systemic risk on the institution.

However, if lending management tightens risk governance by elevating the threshold to $t = 0.55$, Applicant 1 faces outright rejection ($P = 0.48 < 0.55$), while Applicant 2 is approved ($P = 0.52 >= 0.55$). At $t = 0.55$, DP drops by 12.82% to 0.8878, and EO contracts by 13.19% to 0.8750, yielding a Trade-Off Cost ratio of 1.03. Tightening underwriting thresholds disproportionately excludes younger borrowers who possess viable repayment capacities, denying them foundational credit opportunities while older applicants clear the elevated hurdle. A numerical threshold shift is never merely mathematical; it fundamentally alters socioeconomic mobility and individual livelihoods.

Question 7: Defending a Specific Threshold (t* = 0.35)

The most defensible decision threshold identified across the empirical sweep is t* = 0.35. Three concrete empirical findings from the 300-row test cohort justify this operational selection. First, t = 0.35 maximizes overall predictive accuracy across the entire threshold grid, reaching 75.00% by correctly classifying 225 of the 300 test applicants. This outperforms the conventional 0.50 cutoff, which yields 74.00% accuracy. Second, t = 0.35 nearly eliminates demographic disparity between age groups, producing a Demographic Parity ratio of 1.0184 (89.74% young approval versus 88.12% older approval), while maintaining Equalized Odds at 1.0080 and Predictive Parity at 1.0256. Third, it guarantees exceptional credit access for viable borrowers, achieving True Positive Rates of 96.43% for young applicants and 95.05% for older applicants.

The strongest counter-argument against t* = 0.35 arises from institutional credit risk and default loss exposure. A credit risk officer would rightly object that an inclusive 0.35 cutoff tolerates elevated False Positive Rates: 72.73% for young applicants (approving 8 of 11 young defaulters) and 72.15% for older applicants (approving 57 of 79 older defaulters). In retail banking, approving applicants who subsequently default incurs direct charge-offs and administrative collection expenses.

This critique is commercially valid, as expanding inclusion at lower thresholds inherently inflates bad debt exposure. However, an institution can responsibly implement t* = 0.35 by deploying compensating credit controls. Rather than rejecting marginally scoring applicants outright, the bank can approve loans under structured risk mitigations, such as capped credit limits, shorter amortization schedules, or mandatory direct-debit repayment terms. By combining the 0.35 threshold with targeted exposure limits, the bank captures peak classification accuracy and generational equity without jeopardizing regulatory solvency.

Question 8: Demographic Parity vs. Equalized Odds

For commercial credit underwriting, Equalized Odds (EO) should be prioritized over Demographic Parity (DP). A bank loan is a legally binding contract that requires capital repayment, meaning that demonstrated financial capacity and creditworthiness must govern lending decisions. As established in the Part 2.4 threshold audit, DP and EO measure fundamentally different institutional principles. While an inclusive cutoff of t = 0.35 simultaneously satisfies both metrics (DP = 1.0184, EO = 1.0080, accuracy = 75.00%), a conservative operating threshold such as t = 0.55 forces a sharp structural divergence, causing DP to drop to 0.8878 while EO remains more stable at 0.8750.

Within the German Credit dataset, younger applicants possess shorter employment tenures and smaller accumulated savings balances. Consequently, under a strict t = 0.55 cutoff, their empirical approval rate is 66.67%, compared to 75.10% for older applicants. Demographic Parity mandates equal acceptance rates across groups regardless of underlying risk distributions. Enforcing strict DP would compel the institution to artificially lower approval hurdles for high-risk younger applicants, extending loans that borrowers cannot afford and precipitating personal default, distress, and institutional credit losses.

In contrast, Equalized Odds mandates equal accuracy across groups: qualified young applicants receive the same high True Positive Rate as qualified older applicants (85.71% vs. 75.00% at t = 0.55), while error burdens remain equitably distributed. Prioritizing EO over DP acknowledges underlying demographic differences in financial maturity while ensuring that similarly creditworthy individuals receive equal treatment. Sound banking requires evaluating repayment capacity objectively; genuine fairness is achieved by guaranteeing equal opportunity for qualified borrowers rather than enforcing statistical quotas that compromise financial sustainability.

Part 4: Deployment Governance Reflection

> *"If a policymaker asked you: 'Just give me one number: should we deploy this model or not?', what would you say, and what is the limitation of that 'one number' answer that you discovered yourself through your analysis?"*

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1. Direct Recommendation

I recommend conditional deployment with mandatory human review. While the model achieves a seemingly respectable 74.67% test accuracy, that single summary metric conceals an 8.67% confident error rate and severe performance disparities across demographic subgroups. Automated scoring must never operate autonomously on high-stakes lending decisions without structural human safeguards.

2. Why a Single Number Fails

First, an aggregate accuracy score masks deep divergences across demographic groups. In the Simpson's Paradox audit, overall accuracy remained stable, yet underlying feature weights completely flipped across genders. Address tenure exhibited a negative weight overall (beta = -0.0194) but flipped positive for men (beta = +0.0304). Similarly, loan duration predicted defaults strongly for men ($p = 0.002$) while showing no statistical relationship for women ($p = 0.509$), and applicant age strongly protected female borrowers ($p = 0.004$) while having no meaningful effect for men ($p = 0.431$). A single summary number conceals these conflicting patterns, imposing a uniform formula upon groups with fundamentally distinct financial dynamics.

Second, selecting a decision cutoff is an institutional policy decision rather than a neutral mathematical fact. An aggregate metric implicitly assumes that all classification errors carry equal societal and financial costs. A cutoff of $t = 0.35$ maximizes accuracy at 75.00% and balances approval rates (Demographic Parity [DP] = 1.0184), yet tolerates a 72.73% false positive rate among young applicants, exposing the bank to substantial unpaid debt. Conversely, a strict cutoff of $t = 0.55$ shields the portfolio from defaults but rejects 33.33% of creditworthy young borrowers. Furthermore, mathematical impossibility theorems prove that no scoring system can satisfy Demographic Parity, Equalized Odds, and Predictive Parity simultaneously when baseline repayment rates differ across groups. Finally, aggregate scores conceal fragility from omitted variables; dropping checking accounts caused savings weights to surge by +107.30% and employment weights to flip by +260.27%, demonstrating how clean headline metrics can mask borrowed correlations.

3. Supporting Evidence from the Analysis

The empirical stress-tests ground these governance boundaries in concrete metrics. First, linear and logistic architectures disagreed on 82 test cases (a 27.33% mismatch rate). Second, the model generated 26 high-confidence classification errors (8.67% error rate); 45.45% of false positives lacked checking accounts yet secured loans averaging 4,024 Deutsche Mark. Third, omitting checking status produced a +260.27% coefficient swing in employment tenure. Finally, threshold sweeps demonstrated a 1.03 trade-off cost ratio between demographic parity and equalized odds.

4. Key Takeaways for Deployment

Responsible credit underwriting cannot rely on a single performance score. Sustainable deployment demands transparent policy choices regarding solvency versus generational equity, ongoing subgroup residual audits across branch networks, and accessible human appeals for adversely affected borrowers.